Manuel Leite Pinho

Nationality: Portuguese | Phone: (+351) 926265592 (Mobile) | Email address: leite.pinho.manuel@gmail.com | Address: Frankfurt am Main, Germany (Work)

About me

I am a chemical engineer with 4 years experience across process engineering, digital transformation, and research. I enjoy an active lifestyle, I train endurance sports like running or hiking, and enjoy cooking, baking, and hosting board game nights.

Work experience

Process Engineer | Air Liquide | 01/12/2024 - Current | Frankfurt am Main, Germany

- Developing dynamic simulations with **Aspen HYSYS Dynamics** .
- Creating a **PFD** generator tool using draw.io and leading internal training sessions.
- Designing simplified **P&IDs** for clients/suppliers.
- Simulating in **Aspen Plus** the CO2 to Methanol process.
- Executing dynamic plots with **Python** from raw plant data for a **Root Cause Analysis**.

Business Analyst / Project Manager | Galp | 01/07/2023 - 30/11/2024 | Lisbon, Portugal

- Managed technology and digitalization projects across Oil & Gas extraction and refining, including setting up a drilling control room.
- Executed end-to-end process analysis and optimization using SAP Signavio to identify and streamline inefficiencies.

Project Manager | Galp | 03/05/2022 - 30/06/2023 | Lisbon, Portugal

- Project management - SAP S/4HANA support and implementation in different geographies (Portugal, Spain, and Cabo Verde).
- Set-up an automated tool in Power BI and Excel to internaly assist ticketing classification.

Research Engineer | CERENA - Centro de Recursos Naturais e Ambiente | 01/02/2022 - 01/05/2022 | Lisbon, Portugal

- Modeled and controlled a biodiesel washing column with gPROMS Modelbuilder, Matlab, and Simulink. Post-thesis work.

Production and Manufacturing Intern | The Navigator Company | 15/07/2019 - 15/08/2019 | Vila Velha de Ródão, Portugal

- Calibrated an industrial quality assurance scanning machine.

Education & Training

Integrated Master in Chemical Engineering | Instituto Superior Técnico | 14/09/2015 - 25/11/2021 | Lisbon, Portugal

- Thesis: Dynamic Simulation of Biodiesel Washing Column.
- Optional subjects attended: process control, industrial chemistry and process optimization.

International Summer School | Beihang University | 06/07/2018 - 04/08/2018 | Beijing, China

- Thermodynamics, Refrigeration technology, and Chinese culture.

Skills

General | Microsoft Office | SAP S/4HANA | Microsoft Power BI | Python (computer programming)

Process Engineering | Process Simulation | PFD, P&ID, Plant Layout | scientific modelling | use technical drawing software

Languages | Portuguese: native | English: C1 | Spanish: B1 | German: A1 | French: A1

Certifications

Machine Learning, Modeling, and Simulation: Engineering Problem-Solving in the Age of AI | MIT xPro | 15/12/2025

MATLAB Onramp | MathWorks | 04/10/2025

Power BI Strategy @ Galp | Level III - Advanced | Singularity Digital Enterprise | 22/07/2022

Publications

Rigorous Dynamic Modeling and Simulation of an Industrial Liquid–Liquid Extraction Packed Column for Biodiesel Washing

Manuel L. Pinho, Catarina G. Braz, José F. O. Granjo, and Henrique A. Matos (2023). American Chemical Society Industrial & Engineering Chemistry Research Vol. 62, Issue 39, 16000-16013.

Control of an industrial packed extraction column for biodiesel washing

Manuel L. Pinho, Catarina G. Braz, Henrique A. Matos, Carla I.C. Pinheiro, José F.O. Granjo (2022). Elsevier 32nd European Symposium on Computer Aided Process Engineering (ESCAPE-32) Vol. 51, Pages 1123-1128.